

EviSeq v3: Adaptive Full-Memory Retrieval for Composing Pretrained Language Models

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Abstract

Composing a pretrained source encoder with a pretrained causal decoder requires more than matching hidden dimensions: the decoder must retrieve source content at each generation step while the source and target checkpoints remain in different representation spaces. We present EviSeq v3, an adaptive full-memory retrieval architecture for this setting. A value-anchored bridge keeps every source position, builds a multi-scale source prior, and supplies refined keys with final-state values to copied cross-attention in every decoder layer. A four-head semantic read selects 64-token source regions and then source tokens, while a causal prefix planner tracks learned attention usage and recent regions. Head gains are applied after context normalization, and a tangent, norm-preserving fusion bounds the semantic correction to the decoder hidden state. The grounded copy distribution is marginalized over repeated source token IDs and mixed with the language-model distribution. Training uses only gold-token cross-entropy; no evidence labels, contrastive loss, draft summary, or candidate reranking is used. The current implementation passes 318 local tests, but full PubMed ROUGE for the revised v3 graph is not yet available. We therefore present the architecture and locked evaluation protocol without claiming a generation-quality improvement.

1 Introduction

Abstractive summarization compresses a document into a shorter text that preserves its important content. Scientific summarization makes the source interface particularly demanding: a useful abstract must select relevant findings, preserve relationships and quantities, and remain fluent when the input is much longer than the output. A causal language model can generate fluent text, but fluency does not ensure that the generated facts are supported by the document.

Encoder-decoder models learn a source-to-target interface during pretraining and expose source states to the decoder through attention (Sutskever et al., 2014; Bahdanau et al., 2015; Vaswani et al., 2017). This interface is not always available when models are reused from the current checkpoint ecosystem. An embedding-oriented encoder and a causal decoder may have different hidden widths, coordinate systems, tokenizers, and pretraining objectives. A projection can make their widths compatible, but it does not determine how the decoder should retrieve a long source or how an auxiliary source read should affect generation.

Long-document summarization has addressed source access with hierarchical attention and coverage-like mechanisms (Cohan et al., 2018; See et al., 2017). Those mechanisms provide useful precedents, but composing independently pretrained checkpoints adds two constraints. First, the bridge must preserve a full source memory rather than selecting a reference-dependent subset. Second, a semantic source read should change the language-model state without silently changing the exact-copy distribution or the hidden-state scale that the pretrained vocabulary head expects. These constraints motivate a single graph in which source retrieval, copying, and language-model generation are trained by the same token objective.

We introduce EviSeq v3, whose core component is Adaptive Full-Memory Retrieval (AFMR). AFMR uses a document-, prompt-, and output-budget-conditioned controller to combine final and intermediate encoder states, create a multi-scale source prior, and return both a refined key memory and a final-state value anchor. The Qwen3 decoder reads this memory through one copied cross-attention block in every layer. At the output, a grounded copy head reads aligned source positions, while a separate semantic read operates at four heads of 128 dimensions each. It first assigns mass to 64-token source regions and then dis-

tributes that mass within each region. The default graph lets every semantic head inspect every valid region.

The semantic read is causal with respect to the generated prefix. A learned planner estimates source-region usage from the already observed summary prefix, decreases the preference for regions with accumulated usage, and gives a continuity bias to the most recent region distribution. This state is computed by a parallel cumulative scan during teacher forcing and carried through the generation cache at inference. It is a learned attention-usage estimate, not a factuality verifier or a sentence-level discourse planner.

The current v3 revision makes two targeted changes to the earlier implementation. It removes hard modulo-based ownership of source regions, which could leave heads with singleton choices or no valid region on short sources. It also moves the semantic head gain after context normalization, so reducing a gain changes the semantic correction instead of being mostly canceled by a later normalization. These changes leave the total semantic rank at 512, keep the copied cross-attention path, and retain a norm-preserving fusion with relative RMS cap 0.10. The default cross-attention query gate from the exploratory branch is disabled, and the training objective remains gold-token cross-entropy only.

This paper asks four questions:

- **RQ1:** Can AFMR connect an independently pretrained encoder and causal decoder through one full source memory?
- **RQ2:** Does hierarchical region-then-token reading improve over a flat semantic read at the same total rank?
- **RQ3:** Do unrestricted source-region access, prefix usage penalties, continuity bias, and norm-preserving fusion each contribute under matched training?
- **RQ4:** What quality, length, repetition, and computational trade-offs appear on PubMed and auxiliary summarization settings?

The revised v3 code passes its local integration suite, including causal-prefix, cache, masking, gradient, checkpoint, and distributed checks. It has not yet been trained and scored on the full PubMed test split. We consequently treat the prior internal

PubMed scores as historical context, not as results of EviSeq v3, and define the comparisons required to answer RQ1–RQ4.

Our contributions are as follows:

- We formulate AFMR as a full-memory interface for composing a pretrained embedding encoder with a causal decoder. The bridge separates refined key representations from final-state value representations and conditions source priors on the document, prompt, and output budget.
- We implement a four-head, 512-rank semantic read with unrestricted region access, a causal coverage-and-continuity state, post-normalization head gating, and norm-preserving fusion. The design makes each proposed source-reading choice independently ablatable.
- We specify a leakage-resistant, CE-only evaluation protocol with matched controls, deterministic decoding, ROUGE backend locking, and prediction manifests. The current results section reports what has been verified and marks the missing quality experiments explicitly.

2 Related Work

Encoder-decoder summarization and long inputs. Sequence-to-sequence models and attention-based encoders established the standard separation between source reading and target generation (Sutskever et al., 2014; Bahdanau et al., 2015; Vaswani et al., 2017). T5, BART, and mT5 learn this interface jointly during pretraining (Raffel et al., 2020; Lewis et al., 2020; Xue et al., 2021). Hierarchical attention models source documents at multiple granularities, which is useful for long scientific documents (Cohan et al., 2018). EviSeq studies a different interface problem: the encoder and decoder are independently pretrained, so the source-to-target interface must be learned after composition.

Copying and source coverage. Pointer-generator models mix vocabulary generation with attention-based copying and use coverage to reduce repeated attention to the same source content (See et al., 2017). EviSeq retains the useful exact-copy decomposition but marginalizes all aligned source positions that share a token ID

and uses a separate semantic read for paraphrastic generation. Its planner estimates region usage from the generated prefix. The planner is therefore related to coverage as an attention-control mechanism, but it does not use a coverage loss and does not claim to verify that a fact has been expressed correctly.

Pretrained language models as encoders and composed systems. Decoder-only checkpoints can be adapted for representation learning, as shown by LLM2Vec and Generative Representational Instruction Tuning (BehnamGhader et al., 2024; Muennighoff et al., 2024). LaMaTE places a language-model encoder in a translation system, and Qwen3 Embedding provides an embedding-oriented model family (Luo et al., 2025; Zhang et al., 2025b). T5Gemma2 is a closely related reference for arranging modern language-model components as an encoder-decoder system (Zhang et al., 2025a). These systems motivate checkpoint reuse, whereas EviSeq focuses on the source memory, copy interface, and semantic retrieval graph required when the components are not released as a matched encoder-decoder model.

Distinction from EviSeq v2. The earlier EviSeq v2 draft used sentence-level salience, a prompt-conditioned DualBridge route, and attention-aligned contrastive evidence supervision. EviSeq v3 replaces that graph with AFMR, region-level prefix planning, four 128-dimensional semantic heads, and CE-only training. No evidence cache or reference-derived label is required by the v3 default configuration. The two versions should therefore be compared as separately trained graphs under a locked protocol, not by transferring a checkpoint between them.

3 Method

Setting and overview. Let $x = (x_1, \dots, x_S)$ denote the source document and $y = (y_1, \dots, y_T)$ its target summary. The encoder produces final states $H_0 = E_\phi(x) \in \mathbb{R}^{S \times d_e}$ and a small set of intermediate states. The decoder has hidden size d_d and predicts one target token at a time. Source masks distinguish content positions from encoder padding and prompt positions.

AFMR maps the encoder output to two source memories: a refined key memory $H_k \in \mathbb{R}^{S \times d_d}$ and a final-state value anchor $H_v \in \mathbb{R}^{S \times d_d}$. A source prior $b \in \mathbb{R}^S$ is passed to the decoder’s

copied cross-attention and to the source-side reads. Figure 1 shows the complete graph. Section 3.1 describes the bridge, Section 3.2 describes the decoder and grounded copy path, and Section 3.3 describes hierarchical semantic reading and prefix planning.

3.1 AFMR bridge

Controller and source memories. The bridge needs a compact state that can adapt source retrieval to the input and output setting without reading target tokens. We compute masked means of the encoder final states and the fixed decoder instruction embeddings. The two vectors are projected to a controller of dimension 256 and combined with three scalar features: $\log(1 + n_x)$ for the number of visible source tokens, $\log(1 + K)$ for the output budget, and $K / \max(n_x, 1)$. After RMS normalization, the result is a controller vector z .

The default PubMed bridge exposes the last four encoder depth taps, including the final state. Each tap is RMS-normalized, scored by a source-content head, and combined with controller-dependent router weights. A gated low-rank residual then refines the final state:

$$\begin{aligned} H_r &= H_0 + g_d \Delta_d, \\ H_k &= \Pi(H_r) + G_f \odot \Delta_f, \\ H_v &= \Pi(H_0). \end{aligned} \quad (1)$$

Here Π is an identity map when the encoder and decoder widths match and an orthogonally initialized linear map otherwise. Δ_d and Δ_f are low-rank depth and feature corrections, and g_d and G_f are bounded controller-conditioned gates. The value anchor bypasses these learned depth and feature residuals. The cross-attention key therefore receives an adaptive representation, while the value path and grounded source reads retain the final-state content representation.

Multi-scale source prior. The bridge retains all visible source positions and creates a prior rather than a hard source mask. For each window size $w \in \{32, 128, 512\}$, we mean-pool overlapping content windows with 50% overlap. A controller-conditioned scorer assigns each window a scalar score, centers scores within the scale, and overlap-adds them back to token positions. Learned scale weights combine the available scales. The resulting focus signal is passed through

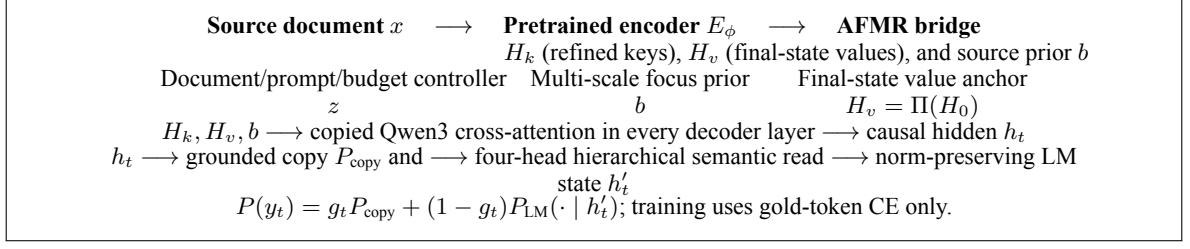


Figure 1: EviSeq v3. AFMR retains the full source, separates key-side refinement from final-state values, and supplies the same source prior to the copied decoder path and semantic read. The prefix planner is causal and uses no reference summary at inference.

a bounded strength gate and a learned temperature in $[0.5, 2.0]$.

For content positions, the prior is centered by its log-mean-exp:

$$b_j = a_j - \left(\log \sum_{s \in \mathcal{C}} \exp(a_s) - \log |\mathcal{C}| \right), \quad (2)$$

$j \in \mathcal{C}.$

where a_j is the bounded multi-scale focus score and $\mathcal{C}$ is the visible content set. Padding and prompt positions receive zero prior. This normalization keeps the mean prior at a stable reference while allowing relative source preferences to affect both cross-attention and semantic reading. It does not select evidence from the reference summary.

3.2 Copied cross-attention and grounded copying

Layer-level source access. Each Qwen3 decoder layer keeps its causal self-attention and then applies a copied cross-attention block before the MLP. The copied block initializes its query, key, value, output, query normalization, key normalization, and source normalization from the corresponding decoder self-attention parameters. For decoder layer ℓ and target position t ,

$$h_{\ell,t}^{\text{cross}} = h_{\ell,t}^{\text{self}} + \gamma_\ell \text{MHA}(h_{\ell,t}^{\text{self}}, H_k, H_v; b), \quad (3)$$

where γ_ℓ is a learned scalar gate initialized to 0.10. The default configuration inserts this block in every decoder layer and disables the exploratory query-dependent cross-head gate. The source prior is an additive attention bias, so content matching can override a weak prior. The source key and value caches are computed once during generation.

Exact copy and token marginalization. The grounded copy head aligns encoder positions to decoder vocabulary tokens using character-offset overlap from the source text and the decoder tokenizer. The alignment procedure uses neither the target summary nor evidence labels. It pools source representations to aligned token IDs, adds a lexical key from the decoder embedding, and computes copy attention:

$$A_{t,j}^{\text{copy}} = \text{softmax}_j \left(\frac{q_t^\top k_j}{\sqrt{d_c}} + b_j \right). \quad (4)$$

If g_t is the learned copy gate, the copy distribution marginalizes repeated source occurrences:

$$P_{\text{copy}}(v | t) = \sum_{j: \text{token}_j=v} A_{t,j}^{\text{copy}}. \quad (5)$$

The semantic read modifies the hidden state used by the vocabulary head, but the copy attention is computed from the pre-fusion hidden state in the same forward pass. This isolates the semantic correction from the copy calculation for fixed hidden states and source state. It does not imply that two independently fine-tuned checkpoints have the same copy distribution, because the shared encoder, decoder, bridge, and copy parameters all receive gradients from the joint objective.

3.3 Hierarchical semantic reading and causal planning

Four unrestricted semantic heads. The semantic read uses total rank $R = 512$ with $H_s = 4$ heads and head dimension $d_s = 128$. It projects the value-anchor source positions to semantic keys and values, then partitions visible content positions into consecutive regions of 64 content tokens. The last region may be shorter. This partition is a data structure for efficient hierarchical reading, not a discourse or sentence segmentation.

For each head h , a query first selects source regions. Let $r(j)$ be the region of source position j , $K_{h,r}$ the mean-pooled region key, and $p_{t,r}$ the planner bias. The region distribution is

$$\pi_{t,h,r} = \text{softmax}_r \left(\frac{q_{t,h}^\top K_{h,r}}{\sqrt{128}} + b_r + p_{t,r} \right). \quad (6)$$

Within each selected region, token attention is normalized separately:

$$\begin{aligned} \alpha_{t,h,j|r} &= \text{softmax}_{j \in r} \left(\frac{q_{t,h}^\top k_{h,j}}{\sqrt{128}} + b_j \right), \\ c_{t,h} &= \sum_r \pi_{t,h,r} \sum_{j \in r} \alpha_{t,h,j|r} v_{h,j}. \end{aligned} \quad (7)$$

The revised default sets `partition_heads=false`, so every head can compare every valid source region. Earlier modulo ownership is retained only as an ablation. Removing this hard assignment matters when a source has fewer regions than heads, or when an important region lies outside the fixed subset assigned to a head. The trade-off is that different heads can attend to the same region; the architecture does not enforce diversity.

Prefix usage planner. The planner estimates usage from the decoder hidden state and region keys. A source-region probability computed from the observed decoder position gives a usage increment only when the position belongs to the summary prefix, not when it is part of the instruction or padding. For region r and prefix position t ,

$$\begin{aligned} u_{t,r} &= \sigma(w_u^\top \text{RMSNorm}(h_t)) \pi_{t,r}^{\text{plan}} \\ &\quad \times \mathbf{1}[t \in \text{summary prefix}], \\ C_{t,r} &= C_{t-1,r} + u_{t,r}. \end{aligned} \quad (8)$$

The planner also carries the most recent valid region distribution $R_{t,r}$. Its bias is

$$\begin{aligned} p_{t,r} &= -\lambda_c \log \left(1 + \frac{C_{t,r}}{\kappa} \right) \\ &\quad + \lambda_r R_{t,r}. \end{aligned} \quad (9)$$

with coverage and continuity enabled in the default configuration. The coverage strength is initialized at 0.2 with maximum 2.0 and scale $\kappa = 8.0$; continuity uses the same initialization and

maximum. During teacher forcing, all prefix states are computed by a cumulative scan, and the chunked loss receives the corresponding plan slice. During generation, the state is updated with each generated token and selected together with compacted self-attention and source-cache rows.

The planner is causal but weakly supervised: it learns through the final token loss. It estimates where the decoder has attended, not whether a fact has been stated correctly. It can therefore support continuity without being interpreted as factual coverage, sentence planning, or an explicit objective-methods-results-conclusion planner.

3.4 Gated semantic fusion

After the hierarchical read, contexts are concatenated and RMS-normalized once. A query-dependent gain is applied separately to each semantic head:

$$\begin{aligned} g_{t,h} &= 2 \sigma(W_g \text{RMSNorm}(h_t) + b_g), \\ u_t &= \text{concat}_{h=1}^4 (g_{t,h} \bar{c}_{t,h}). \end{aligned} \quad (10)$$

where $\bar{c}_t = \text{RMSNorm}(\text{concat}_h c_{t,h})$ and the gain is applied after this normalization. The global semantic gate reads the query and the normalized, ungated context. A zero-initialized output projection produces a correction δ_t .

The default fusion removes the radial component of the correction, bounds its relative RMS, and returns the fused state to the norm of the original decoder hidden:

$$\begin{aligned} \delta_t^\perp &= \delta_t - h_t \frac{h_t^\top \delta_t}{\|h_t\|_2^2}, \\ c_t &= 0.10 \text{ RMS}(h_t), \\ \hat{\delta}_t &= \delta_t^\perp \frac{c_t}{\sqrt{c_t^2 + \text{RMS}(\delta_t^\perp)^2 + \epsilon}}, \\ h'_t &= (h_t + \hat{\delta}_t) \frac{\|h_t\|_2}{\|h_t + \hat{\delta}_t\|_2}. \end{aligned} \quad (11)$$

The zero-hidden case falls back to h_t . This operation preserves hidden-state magnitude up to numerical precision while allowing its direction, and therefore its vocabulary logits, to change. Applying gains after normalization is important for the intended attenuation behavior: a common gain reduction changes the correction before the final fusion instead of being canceled by a second context normalization. The pre-normalization gate and the older residual fusion are explicit ablations.

3.5 Training and inference

For each target position, the model mixes the vocabulary distribution from h'_t with the exact-copy distribution:

$$P(y_t = v \mid y_{<t}, x) = (1 - g_t)P_{\text{LM}}(v \mid h'_t) + g_t P_{\text{copy}}(v \mid t). \quad (12)$$

The only training objective is gold-token cross-entropy,

$$\mathcal{L}_{\text{CE}} = - \sum_{t=1}^T \log P(y_t \mid y_{<t}, x). \quad (13)$$

There is no evidence mining, contrastive loss, R-Drop, NEFTune, teacher model, pseudo-reference, candidate generation, or self-improvement in the default v3 graph. The planner receives the teacher-forced prefix through the decoder input, but not the future target token. At inference, it receives only the generated prefix.

Training uses one encoder forward and one teacher-forced decoder forward. The interface warm-up stage freezes the backbone and trains the bridge, copied cross-attention, grounded copy, semantic read, and planner. Full fine-tuning then updates the encoder and decoder together. During generation, source keys, values, semantic caches, and region caches are prepared once; the planner history and self-attention cache advance with each generated token. A finished batch row is compacted together with all of these states.

4 Experiments

Evaluation status. The revised v3 graph was completed and tested locally on 2026-09-09. Full PubMed training, CUDA throughput measurement, Perl ROUGE-1.5.5 scoring, and matched v3 ablations have not yet been run in the current environment. This section therefore separates implementation evidence from generation-quality evidence. A test-suite pass confirms specified invariants on fixtures; it does not establish a ROUGE gain, factuality improvement, or superiority to T5Gemma2.

Datasets and preprocessing. The main task is PubMed abstract summarization. The prepared snapshot contains 116,699 training records and 6,657 test records; the validation count is stored in the data manifest and is not hard-coded here because snapshots can differ. Each record contains

Variant	Semantic configuration and isolated question
V2 control	1×128 flat read with residual fusion; same-rank control
Free 4×32	Four heads, total rank 128; head count at fixed rank
Free 4×128	Four heads, total rank 512; rank and head width
Pre-review v3	4×128 with partition and pre-norm gate; earlier graph
No partition	4×128, unrestricted, pre-norm gate; region access
No coverage	Revised graph without coverage bias; prefix usage penalty
No continuity	Revised graph without recent-region bias; prefix continuity
Revised v3	4×128, unrestricted, post-norm gate; full graph
Residual fusion	Revised read with residual fusion; hidden-state control

Table 1: Planned matched PubMed ablations. Each row requires an independent train-from-pretrained run; disabling a module in a finished checkpoint is only a diagnostic. Results are pending.

a source list and a summary list. The source is serialized as one sentence per line, with a 4,096-token source limit and a 512-token target limit. PubMed uses the PPLX-Embed v1 0.6B encoder and Qwen3-0.6B decoder configuration.

WikiLingua is an auxiliary cross-domain setting with 13,999 training, 1,680 validation, and 3,901 test records in the current snapshot. Its source and target limits are 3,072 and 384 tokens. CNN/DailyMail is configured with 4,096 source tokens and 256 target tokens. These auxiliary settings are evaluation targets for the same graph, not completed v3 results in this draft. The v3 preparation path uses canonical `id`, `source`, and `target` JSONL records and does not create an evidence-label cache.

Models and matched controls.

The default PubMed model uses `perplexity-ai/pplx-embed-v1-0.6b` as the encoder and `Qwen/Qwen3-0.6B` as the decoder. A Qwen3-Embedding encoder is supported as a paired alternative. The primary ablations keep the pretrained backbones, data snapshot, prompt, source and target limits, optimizer, seed, stages, precision, checkpoint rule, and greedy decoder fixed:

The revised v3 configuration uses `rank=512`, `num_heads=4`, `region_size=64`, `partition_heads=false`, `head_gate_position=post_norm`,

fusion=norm_preserving, and max_relative_rms=0.10. The default cross query-head gate is disabled. The source prior uses window sizes 32, 128, and 512 with 50% overlap. No run may resume a checkpoint produced by a graph with a different semantic rank, head count, partition setting, fusion, or gate position.

Optimization and decoding. The PubMed recipe uses FP32 parameters, gradients, and AdamW states with BF16 CUDA autocast. It runs one interface warm-up epoch followed by three full-fine-tuning epochs, global batch size 96, gradient clipping at 1.0, and random seed 42. The single-GPU equivalent is batch 48 with accumulation 2; the two-GPU runner uses batch 48 per GPU with accumulation 1. Warm-up bridge and copied-cross-attention learning rates are 10^{-4} ; full-stage encoder and decoder rates are 3×10^{-5} , and bridge and copied-cross-attention rates are 5×10^{-5} . Linear decay is applied separately in the two stages. The evaluator uses `last.pt` for the locked benchmark configuration.

All benchmark generations use greedy decoding with one beam and sampling disabled. PubMed uses 512 maximum new tokens, a 32-token minimum, repetition penalty 1.05, and no repeated 3-gram. The generation API retains temperature and top- p for exploratory candidate sampling, but the benchmark never calls that path.

Metrics and provenance. The primary metrics are ROUGE-1, ROUGE-2, and ROUGE-L F1. The evaluator can report Python ROUGE 1.0.0 as a diagnostic backend and can invoke a fixed Perl ROUGE-1.5.5 wrapper for the paper score. Scores from different backends will not be combined. We also record prediction length, reference length, length ratio, empty and truncated predictions, repeated-trigram rate, unique prediction rate, and source-copy rate. These diagnostics distinguish a metric change caused by output length or repetition from a change in source use.

Every v3 prediction file has a manifest containing checkpoint and dataset hashes, inference configuration, tokenizer identity and special IDs, package and library fingerprints, and device type. The evaluator verifies this manifest before resuming or returning metrics, including for a complete prediction file. A mismatch requires a new output path. This check is necessary because a prediction file with correct IDs and references can still con-

Reported run	R-1	R-2	R-L
EviSeq new	49.626	21.901	45.895
EviSeq update v1	49.352	21.734	45.598
EviSeq update v2	49.488	21.953	45.776
Revised EviSeq v3	<i>pending</i>	<i>pending</i>	<i>pending</i>

Table 2: Historical internal PubMed ROUGE values and the unmeasured revised v3 row. The historical rows are context only; scorer and training-protocol differences prevent a causal comparison.

tain predictions from another checkpoint.

5 Results and Discussion

Implementation validation. The revised package passes 318 local tests. The suite covers the hierarchy oracle, unrestricted and partitioned routing, one-to-four source regions, prefix-only planning, prompt and padding exclusion, dense and chunked CE parity, FP32 and BF16 gradients, norm and fusion bounds, copy isolation for fixed hidden states, cache compaction, checkpoint compatibility, evaluation provenance, and two-process Gloo distributed execution. A tiny two-stage smoke run saved and reloaded a checkpoint, generated two examples, and reduced fixture CE from 4.490708 to 4.476885. These observations support implementation consistency on controlled fixtures. They do not measure PubMed summarization quality.

Historical PubMed context. Table 2 records scores reported for earlier internal EviSeq runs. They motivate a controlled v3 study but are not v3 measurements. The reports do not provide a single reconstructed protocol that is sufficient to treat the rows as a clean architecture-only comparison, so we do not claim that one row is statistically better than another.

The missing row is substantive rather than cosmetic. The default v3 changes both the source-region support available to semantic heads and the location of their gain relative to normalization. It also changes the number of semantic dimensions from the earlier 1×128 control to 4×128 . A full test score is therefore needed before attributing any change to unrestricted routing, coverage, continuity, or fusion.

Planned analysis. The locked comparison will report all three ROUGE metrics under one scorer, paired bootstrap intervals over identical examples,

and the diagnostics listed in Section 4. The ablation table is designed to answer the four questions without changing multiple causes at once. The no-partition and pre-review rows isolate the two revisions made after review. The no-coverage, no-continuity, and residual-fusion rows isolate the planner and fusion choices. The flat and free-head controls separate the effect of hierarchy from total semantic rank.

The analysis will also report the number of valid source regions, planner coverage and recent-region distributions, semantic residual RMS relative to the decoder hidden, and the fraction of examples with an empty or truncated source. These measurements are intended to test the proposed mechanism. Concentrated attention or a bounded residual alone will not be treated as evidence of factuality or improved summaries.

6 Limitations

The strongest limitation of this draft is that revised v3 has no full PubMed generation result. The architecture and local tests cannot establish an improvement over EviSeq v2, T5Gemma2, a decoder-only baseline, or any other system. The planned ablations and scorer-locked test evaluation are required before such claims are appropriate.

The current method also has structural limits. The 64-token regions are position-based and may cut sentences or facts. Unrestricted heads can learn overlapping reads, and the planner estimates attention usage rather than verified fact coverage. The value anchor and refined key memory separate roles but add source projections, cache storage, and computation. The implementation has not yet reported CUDA throughput or peak memory at the production batch and source lengths.

Finally, the evidence scope is narrow. The main configuration targets English biomedical summarization and uses one encoder-decoder composition. ROUGE measures reference overlap and does not by itself measure factual consistency. The source-to-token alignment is conservative around truncation boundaries, which may reduce copy opportunities. Multiple seeds, human factuality assessment, and adapter or parameter-efficient training comparisons remain future work.

7 Conclusion

EviSeq v3 defines a full-memory interface for composing a pretrained source encoder with a causal Qwen3 decoder. AFMR returns refined keys, final-state values, and a multi-scale source prior. The decoder reads this memory at every layer, while a grounded copy head and a separate four-head semantic read provide complementary source access. Hierarchical region reading, causal prefix usage, post-normalization head gains, and norm-preserving fusion are implemented as separately testable choices under a single gold-token CE objective.

The current evidence supports the implementation contract, not a quality claim: 318 local tests pass and the smoke pipeline trains, saves, loads, and generates on fixtures, while the revised v3 PubMed ROUGE row remains pending. The next decisive experiment is a train-from-pretrained, scorer-locked comparison of the matched controls in Table 1. Until that experiment is complete, EviSeq v3 should be presented as a reproducible architecture and evaluation plan rather than as a demonstrated improvement.

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A Configuration and Reproducibility

The default PubMed v3 configuration is listed below to make the graph and the planned comparison reproducible.

Component	Configuration
Encoder	PPLX-Embed v1 0.6B
Decoder	Qwen3-0.6B
AFMR	Value anchor, controller dimension 256, four depth taps
Depth correction	Rank 128, gate init 0.02, maximum 0.15
Feature correction	Rank 256, gate init 0.02, maximum 0.20
Focus prior	Windows 32/128/512, overlap 0.5, strength init 0.10
Focus temperature	Init 1.0, range [0.5, 2.0]
Cross-attention	Copied in every layer, scalar gate init 0.10
Semantic read	Hierarchical coverage, rank 512, four heads of 128
Source regions	64 content tokens, unrestricted heads
Planner	Coverage and continuity on, init 0.2, maximum 2.0, scale 8.0
Semantic gate	Post-normalization head gate, init 1.0 gain
Fusion	Tangent, norm-preserving, relative RMS cap 0.10
Objective	Gold-token mixture cross-entropy only
Training	One warm-up plus three full epochs, global batch 96, seed 42
Precision	FP32 parameters/gradients, BF16 CUDA autocast
Decoding	Greedy, one beam, sampling disabled

The default data prompts are biomedical summarization instructions supplied to the encoder and decoder. The decoder prefix is `Abstract:`. The evaluator uses `last.pt` and creates a new prediction output when any checkpoint, split, tokenizer, inference setting, or implementation fingerprint changes. The v3 package does not resume a checkpoint across a graph-spec mismatch, even when the tensor shapes happen to agree.

B Source Alignment and Evaluation Protocol

The copy alignment is computed from source character offsets and decoder tokenizer offsets. A source token contributes to an aligned decoder token only when its non-space span is covered by the overlap rule. Alignment weights are normalized per decoder token, and tokens with no valid content overlap are masked. The target summary is not used to construct this state. This prevents the exact-copy path from receiving reference-derived evidence labels.

For training, the decoder receives a teacher-forced input sequence containing the fixed prompt and the observed target prefix. The planner mask excludes prompt and padding positions. Its full prefix state is computed once by cumulative sum, then sliced for chunked CE. For generation, the planner history starts empty, advances only on generated positions, and is selected with the same rows as the self-attention, cross-attention, and

source caches when finished examples are compacted.

The paper score will be computed by one fixed Perl ROUGE-1.5.5 wrapper with the same tokenization, stemming, file naming, and detokenization choices for every model. Python ROUGE is retained for diagnostics and will be labeled separately. Test predictions will be generated once for the locked configuration. Validation is used for graph selection; the test split is not used for repeated gate, head, or fusion tuning.

C Adversarial Draft Self-Review

This checklist is part of the revision record and should be removed from a submission copy.

Contribution (new experiment needed).

AFMR provides a concrete composition graph and isolates source-reading choices, but the current draft has no v3 quality result and cannot yet establish a generation-quality contribution.

Clarity (pass for current scope). The paper defines the two source memories, the planner state, the semantic read, the fusion operation, and the CE objective. The distinction between learned attention usage and factual coverage is stated explicitly.

Experiments (new runs needed). The local test suite checks implementation invariants only. PubMed ROUGE, strong baselines, auxiliary datasets, and multiple seeds remain to be run.

Evaluation (new runs needed). The ablation matrix, scorer lock, paired uncertainty, length and repetition diagnostics, and provenance manifest are specified, but their final measurements are missing.

Soundness (pass with limits). The revised graph has causal prefix handling, empty-source fallback, cache compaction, and checkpoint guards. The unresolved costs are semantic cache size, unmeasured CUDA performance, position-based regions, and possible head overlap.